

Edge AI Patent Assessment

Rendered with default font. Install NotoSansCJK for improved CJK coverage.

Total Score: 66.45

Label: X

Flags: FTO_risk

API calls 0 (cached 0, batches 0); Embedding 0 snippets (~0 tokens / cap=25000); Deterministic metrics

Synthesis

Problem: identify enforceable claims for edge AI execution deployments. Solution: independent claims highlight configurations such as A device comprising an on-device processor configured to execute a quantized neural network in real-time within an opera... Evidence: TRL estimate 8.5 with 0 forward citations supports readiness.

API Metrics

Metric (Status) — Value / Year / Source / URL

CITATIONS_TOTAL (budget_exceeded)

Value: N/A Year: 2000
Source: NBER Working Paper 7741 [NBER-W7741]
URL: <https://www.nber.org/papers/w7741>

FAMILY_SIZE (budget_exceeded)

Value: N/A Year: 2024
Source: WIPO IP Indicators [WIPO-2024]
URL: <https://www.wipo.int/web-publications/world-intellectual-property-indicators-2024-highlights/en/patents-highlights.htm>

COUNTRIES (budget_exceeded)

Value: N/A Year: 2024
Source: WIPO IP Indicators [WIPO-2024]
URL: <https://www.wipo.int/web-publications/world-intellectual-property-indicators-2024-highlights/en/patents-highlights.htm>

LITIGATION_FLAG (budget_exceeded)

Value: N/A Year: 2012
Source: USPTO Official Gazette [USPTO-OG-2012-52]
URL: <https://www.uspto.gov/web/offices/com/sol/og/2012/week52/TOCCN/item-80.htm>

RENEWAL_STATUS (budget_exceeded)

Value: N/A Year: 2012
Source: USPTO Maintenance Fee [USPTO-FEE]
URL: <https://www.uspto.gov/web/offices/com/sol/og/2012/week52/TOCCN/item-80.htm>

SEP_INDICATION (budget_exceeded)

Value: N/A Year: 2022
Source: ETSI SEP List [ETSI-SR-000-314]
URL: https://www.etsi.org/deliver/etsi_sr/000300_000399/000314/02.32.01_60/sr_000314v023201p.pdf

HHI (budget_exceeded)

Value: N/A Year: 2024
Source: DOJ / FTC Guidelines [DOJ-HHI]
URL: <https://www.justice.gov/atr/herfindahl-hirschman-index>

GENERALITY (budget_exceeded)

Value: N/A Year: 2001

Source: NBER Working Paper 8498 [NBER-W8498]

URL: <https://www.nber.org/papers/w8498>

ORIGINALITY (budget_exceeded)

Value: N/A Year: 2000

Source: NBER Working Paper 7741 [NBER-W7741]

URL: <https://www.nber.org/papers/w7741>

AGE_YEARS (budget_exceeded)

Value: N/A Year: 2025

Source: Google Patents [GOOGLE-PATENTS]

URL: <https://patents.google.com/patent/WO2018097365A1>

Score Breakdown

trl	85.00 [Local RAG]
claim	85.00 [N/A]
legal	60.00 [API meta]
family	45.00 [API meta]
foreign	45.00 [API meta]
renewal	40.00 [API meta]
std	30.00 [API meta]
generality	N/A [API meta]
originality	N/A [API meta]
fto	0.00 [API meta]

Evidence - TRL [Local RAG]

Estimated TRL Level: 8.5

1. Example progression: **prototype** -> **pilot** -> field test -> **deployed production** services.
2. An on-device inference apparatus is described, transitioning a **prototype** into a **pilot** deployment with real-time quantized processing for edge AI scenarios.
3. TRL 5 corresponds to validating a software **prototype** in a relevant environment. TRL 6 and TRL 7 involve field test pilots with partially **deployed** systems. TRL 8 and TRL 9 indicate **production-ready** and fully **deployed** software in an operational environment.

Evidence - CLAIMS [Local RAG]

Independent claims: 1, Avg tokens: 177.0

1. A device **comprising** an on-device processor configured to execute a quantized neural network in real-time within an operational environment, the device including telemetry for **prototype** to **pilot** upgrades, adaptive calibration pipelines, secure enclave management, federated aggregation buffers, and a plurality of edge accelerators coordinated by a policy engine that maintains latency thresholds below ten milliseconds while preserving power budgets across mission profiles. The device further comprises memory partitions, compression controllers, wireless radios, fallbacks for disconnected operation, proactive failure detection routines, and orchestrated deployment hooks that enable continuous integration of model updates validated through field trials, **pilot** deployments, and **production** readine

2.

The device of claim 1 **wherein** the processor is configured to execute federated updates on-device, dynamically batching encrypted gradients, reconciling conflict resolution policies, and orchestrating synchronization checkpoints. The device of claim 2 **wherein** the memory subsystem provides at least one secure enclave with attestation pipelines, redundancy failover, and audit telemetry forwarding pathways.

Evidence - API_METRICS [Local RAG]

Evidence - SCORE [Local RAG]